ICML
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On Machine Learning

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# Evaluating Object-Centric Models beyond Object Discovery



Krishnakant Singh [1]

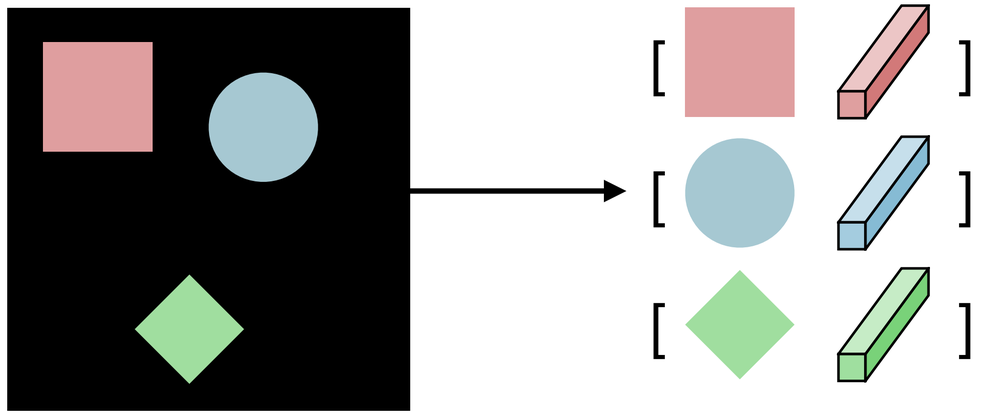

Simone Schaub-Meyer [1,2]


Stefan Roth [1,2]

[1] TU Darmstadt [2] hessian.AI

# Object centric representation learning

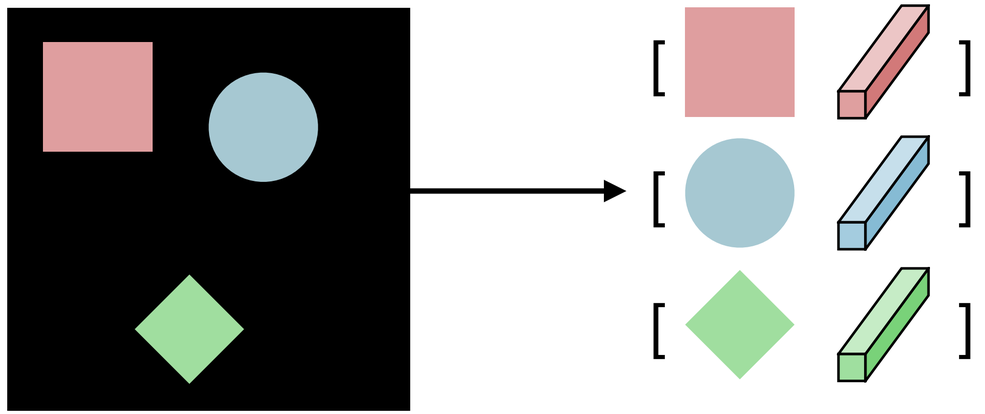
- Learn a dedicated representation for each object in the scene



Slide Adapted from Fei Deng.

# Object centric representation learning

- Learn a dedicated representation for each object in the scene



An ideal slot embedding for an object encodes both

- Position (*Where*)
- Attribute (*What*)

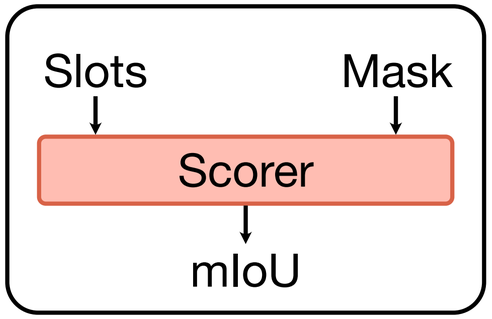
Slide Adapted from Fei Deng.

# Evaluation of OCL methods

- Object Discovery



Input Image Pred. Mask G.T. Mask



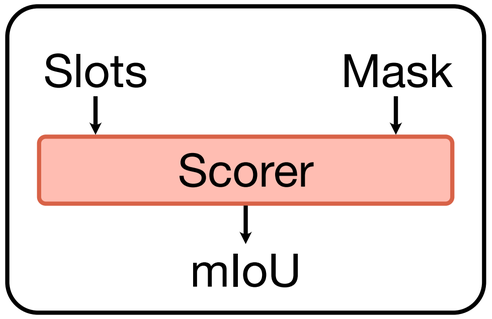

Evaluates only if slot localizes the object

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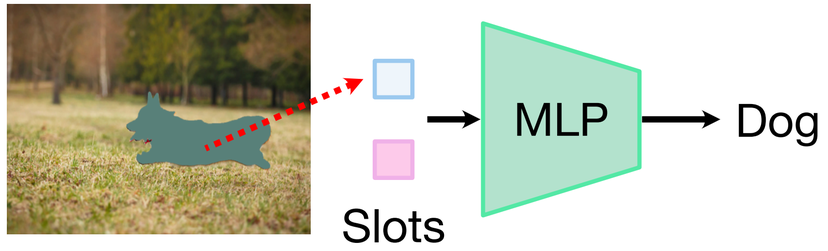


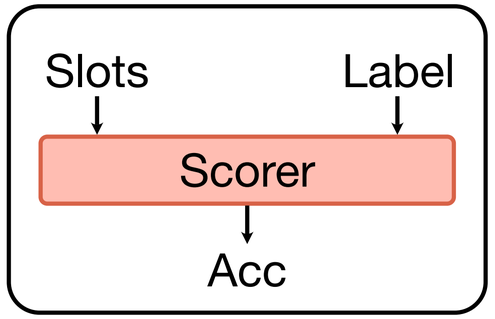
Input Image Pred. Mask G.T. Mask




Evaluates only if slot localizes the object

- Property prediction (Linear Probing)




Not scalable

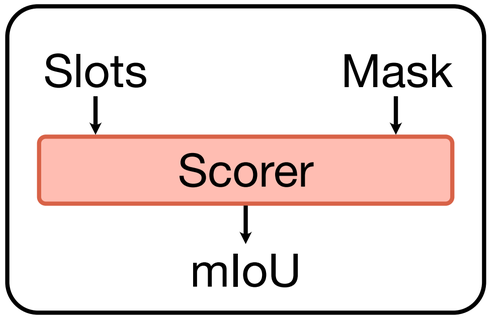
Simple properties

# Evaluation of OCL methods

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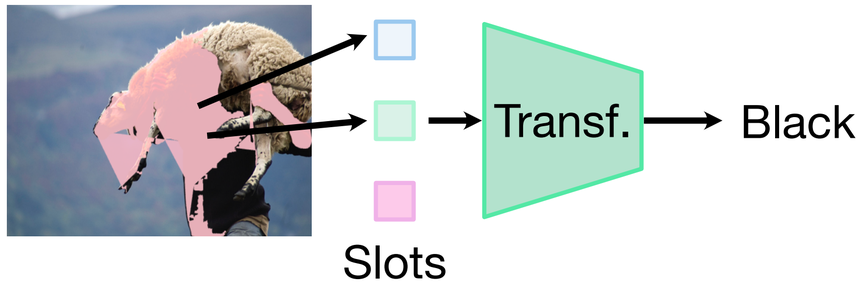


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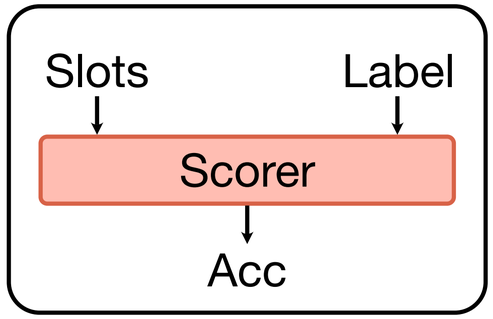



Evaluates only if slot localizes the object

- Property prediction (VQA)




*What color is the t-shirt?*




Not scalable

Simple properties

Mamaghan *et al.,* Exploring the Effectiveness of OCL in VQA: Comparative Insights with Foundation Models, ICLR (2025).

# Diverse goals of OCL

## Out of Distribution Robustness



Shape




Context



Texture

# Diverse goals of OCL

## Out of Distribution Robustness



Shape



Context



Texture

## Natural Adversarial Robustness



Is the squirrel climbing up or down?



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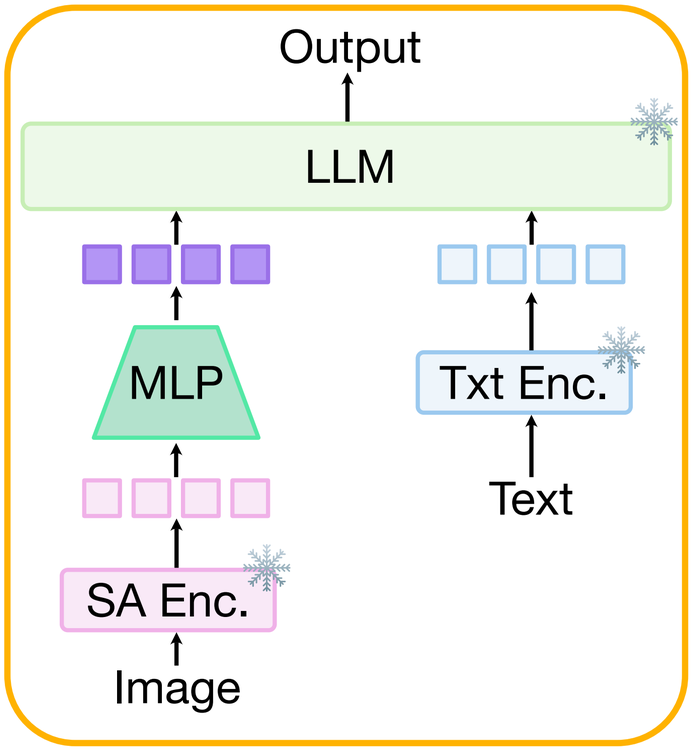
## Counterfactual Reasoning



How many people would be there if 2 were added?

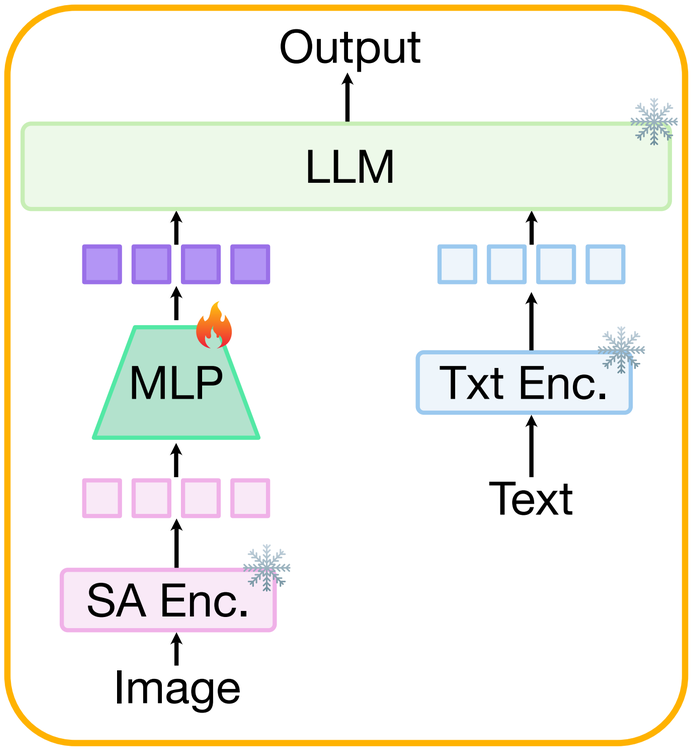
# Towards a more scalable evaluation protocol

- Scalable
- Evaluate diverse properties



# Towards a more scalable evaluation protocol

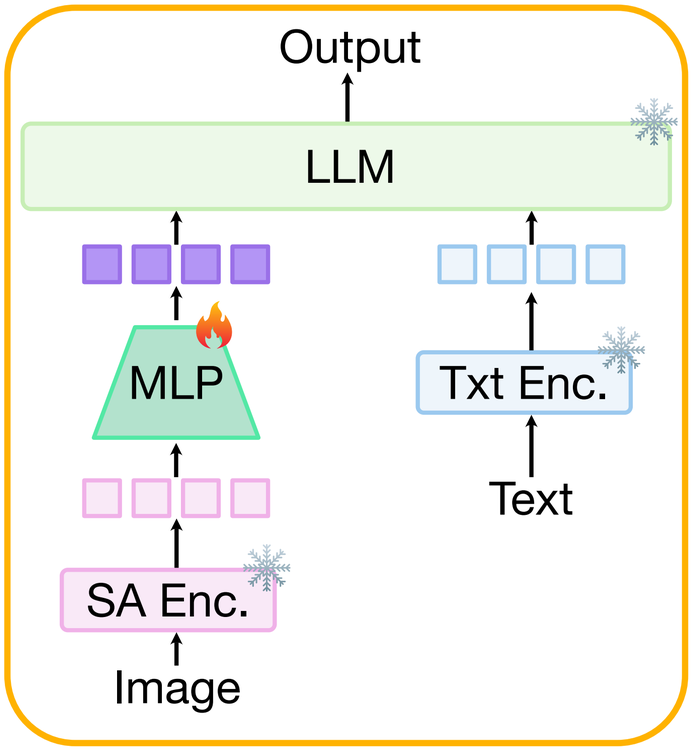
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Alignment Stage

# Towards a more scalable evaluation protocol

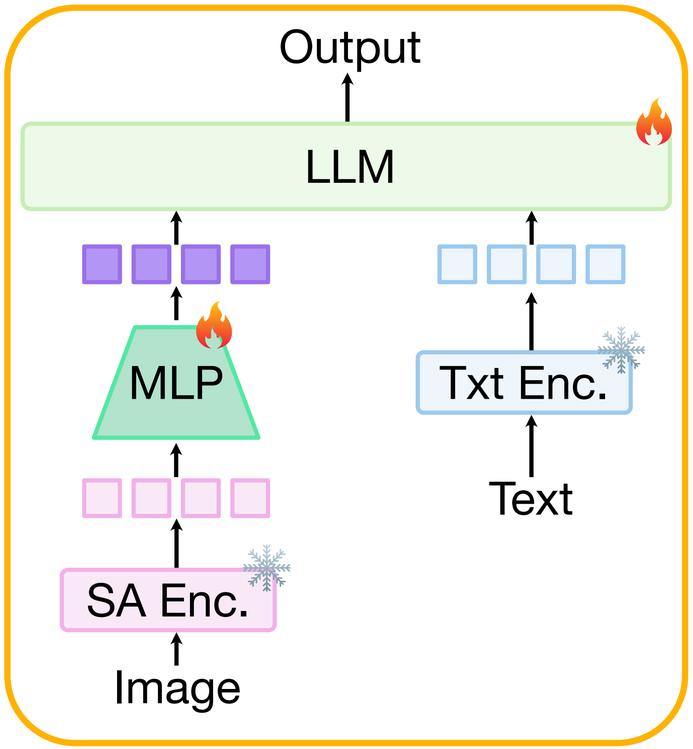
- Scalable
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Instruction Tuning Stage

# Towards a more scalable evaluation protocol

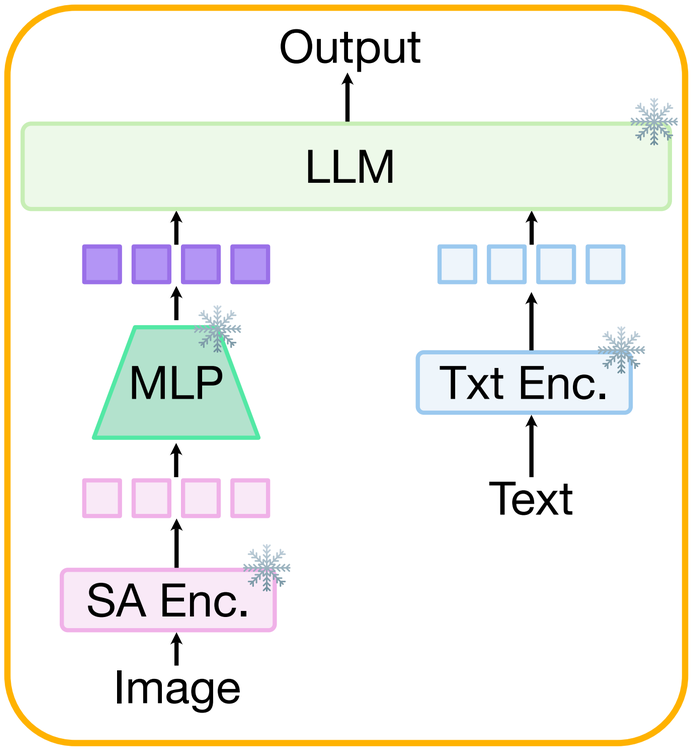
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Instruction Tuning Stage

# Towards a more scalable evaluation protocol

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Evaluation Stage

# Evaluating on VQA

| Model | GQA | POPE | VQAv2 |
|---|---|---|---|
| DINOv2 | 57.77 | 82.01 | 71.15 |

Hudson *et al., GQA: A new dataset for real-world visual reasoning and compositional question answering*, CVPR (2019).

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OCL methods still behind DINOv2


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OCL methods lag behind DINOv2 but are comparable on certain benchmarks


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# Evaluating on VQA

| Model | CVQA (D) | OODCV | SugarC. |
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| Model | CVQA (D) | OODCV | SugarC. |
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| DINOv2 | 36.96 | 58.00 | 82.05 |
| DINOSAURv2 | 34.52 | 53.90 | 75.98 |
| SPOT | 36.35 | 53.34 | 71.65 |
| StableLSD | 38.26 | 52.89 | 72.92 |

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OCL still lags behind DINOv2 on compositional and natural adversarial benchmarks

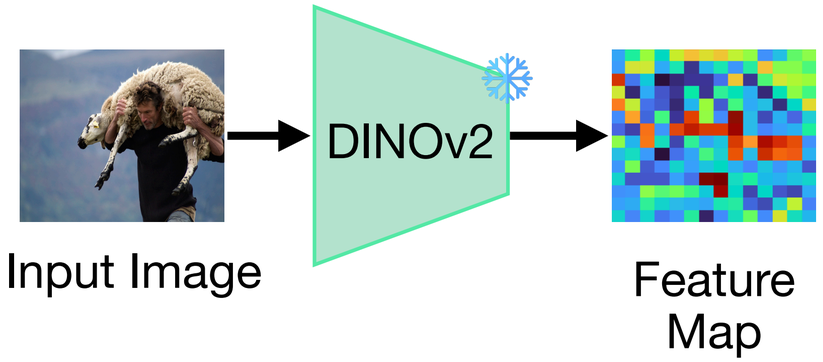
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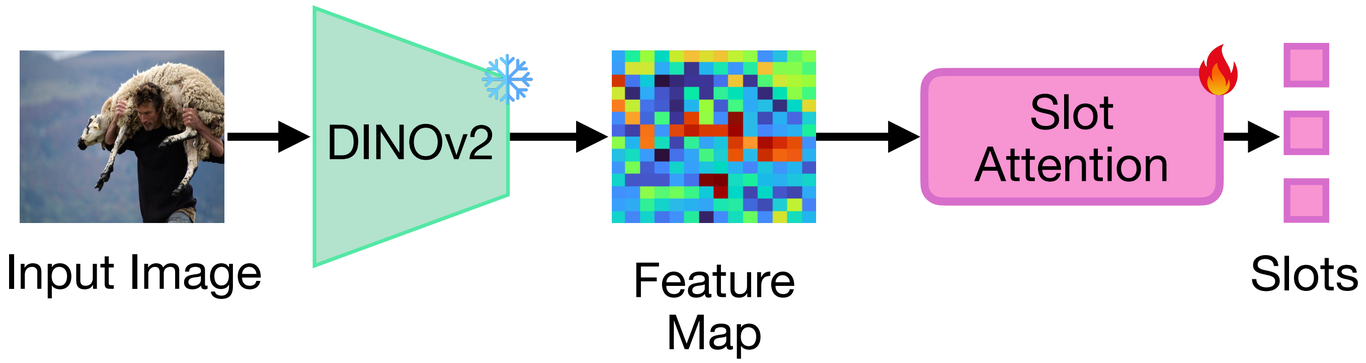
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# Can we improve the results?

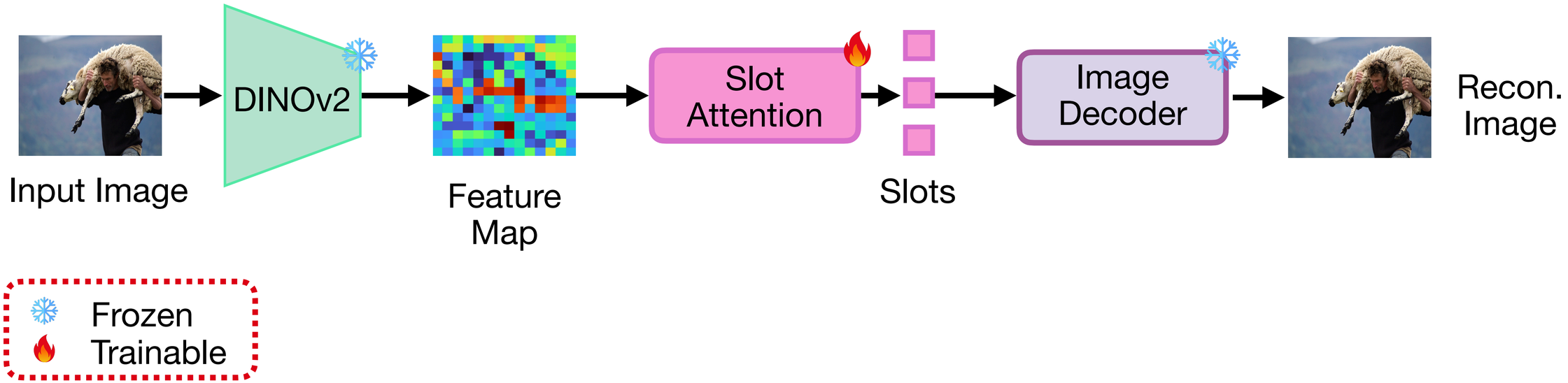
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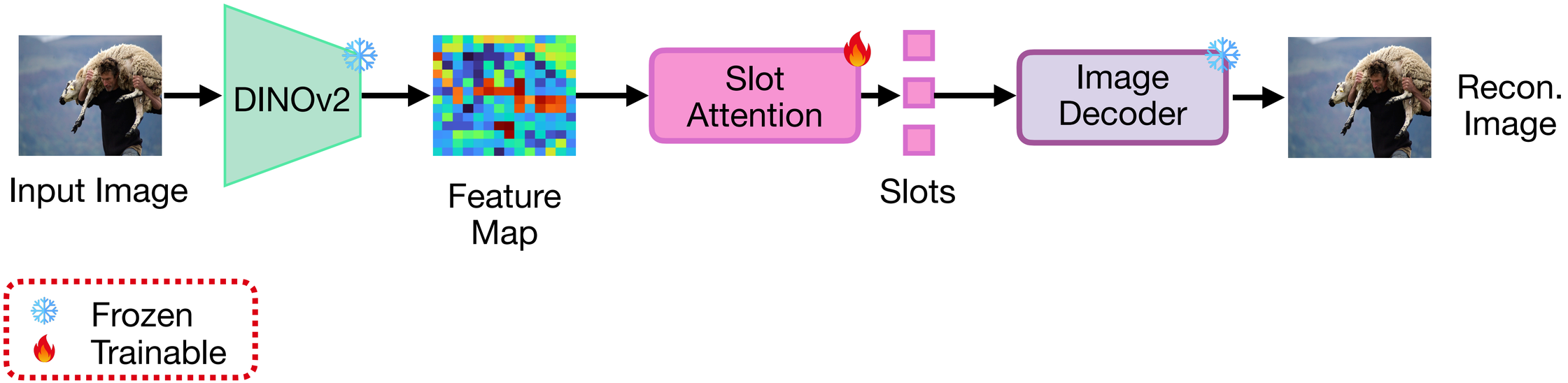
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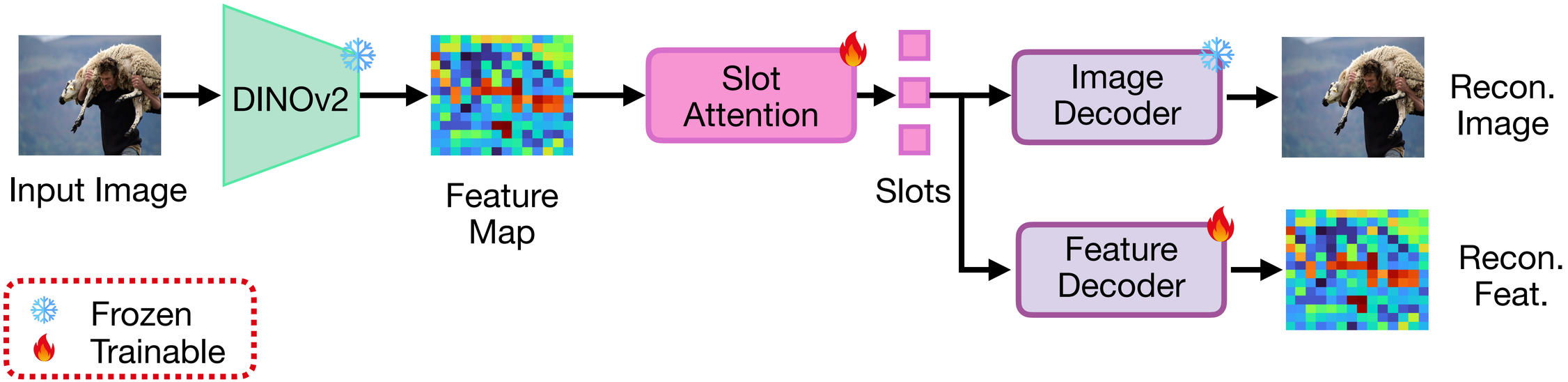
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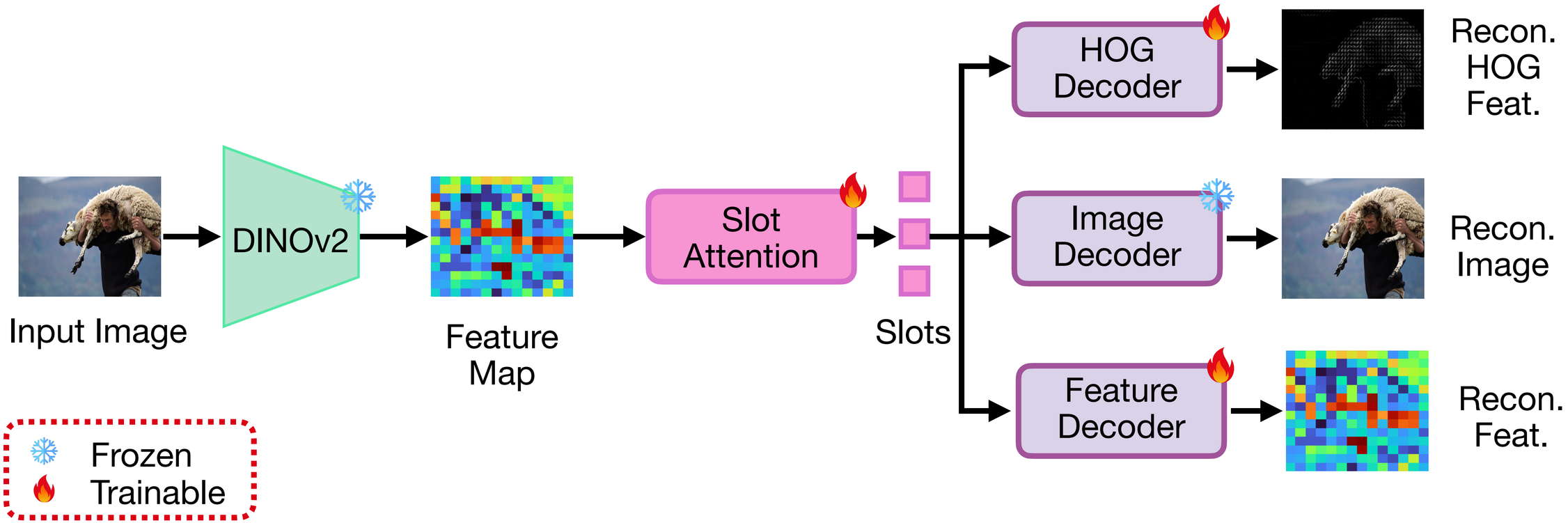
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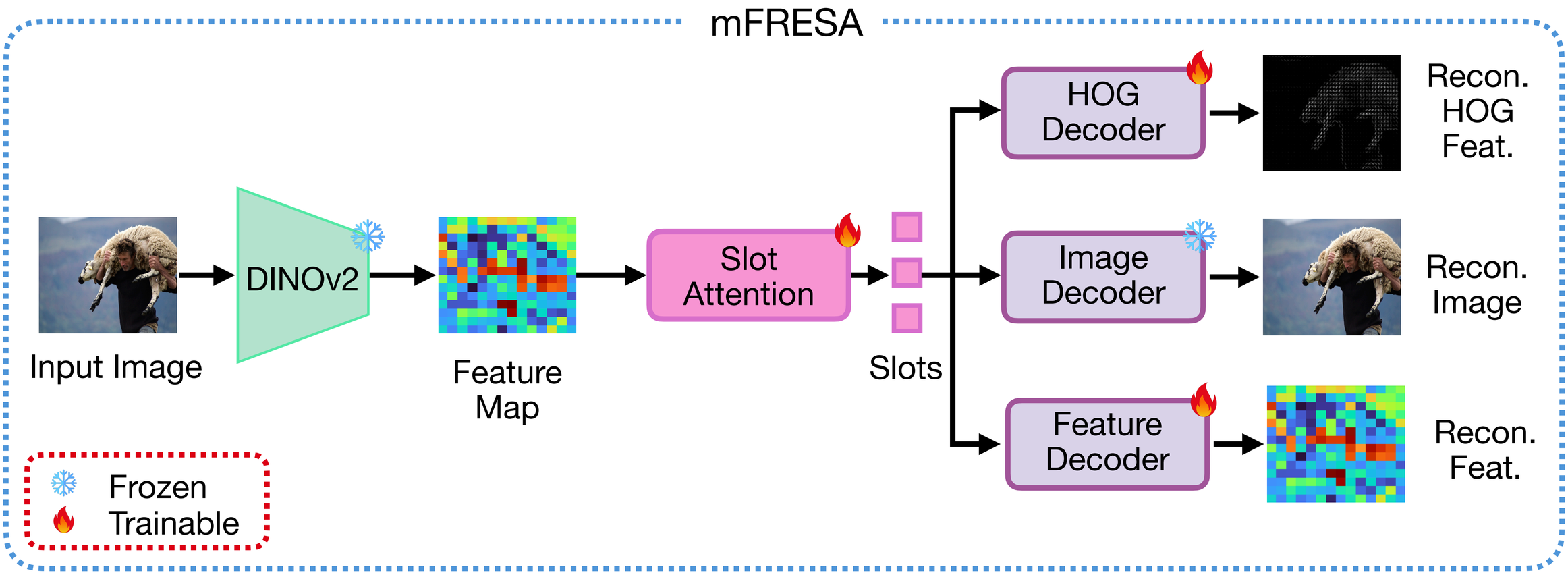
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| mFRESA *(ours)* | **53.90** | **82.12** | **65.58** | 38.09 | **55.57** | **77.27** |

+0.67 +0.36 +1.74 +1.67 +1.29

mFRESA improves performance on most VQA benchmarks

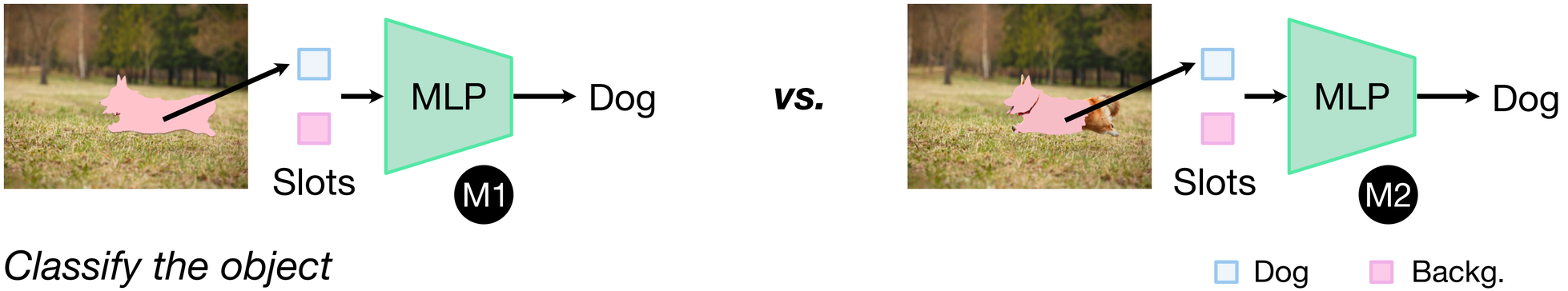
# Issues with disjoint evaluation

- Localization Fragmentation



# Issues with disjoint evaluation

- Localization Fragmentation




- Incoherence between localization and property prediction

| Metric | mIoU | Acc. |
|---|---|---|
| DINOSAURv2 | 47.99 | 66.27 |
| FT-DINOSAUR | **59.09** | 61.05 |
| SPOT | 53.76 | 64.08 |
| mFRESA *(ours)* | 56.92 | **67.58** |

Object discovery is a poor proxy of slot usefulness

# eGQA dataset

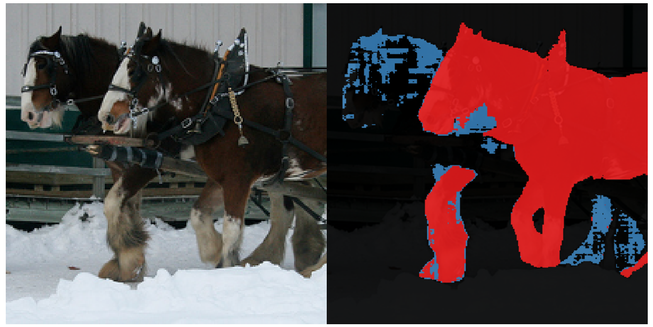
- Joint evaluation requires a dataset with grounding annotations
- We introduce eGQA, 10K visual question answer pairs with grounding annotations

# eGQA dataset

- Joint evaluation requires a dataset with grounding annotations
- We introduce eGQA, 10K visual question answer pairs with grounding annotations



What color is the uniform?
Ans: Red



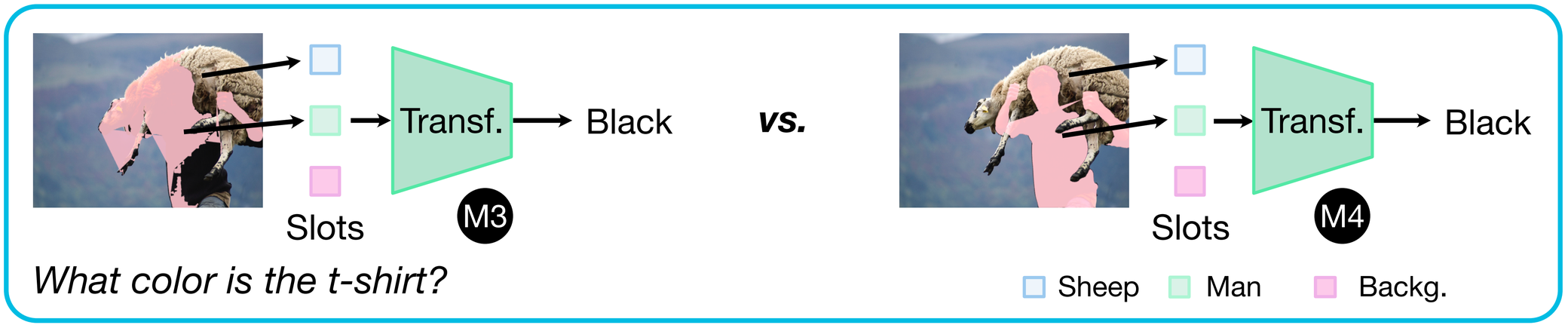
What kind of animal is pulling the cart?
Ans: Horse



Do you see an umbrella above her?
Ans: Yes

# Joint evaluation

- Naïve Fix: Grounded Accuracy
- Representation Fragmentation



# AwGA: Attribute aware Grounded Accuracy

$$\mathrm{AwGA} = \frac{1}{N} \sum_{i=1}^{N} \underbrace{\mathbb{1}(\hat{y} = y)}_{\text{Correct predictions}} \underbrace{\mathrm{mIoU}(\mathrm{TopK}(\mathcal{A}_{\mathrm{pred}}), \mathcal{G}_{\mathrm{GT}})}_{\text{Overlap between the top-k mask. and GT masks}}.$$

# AwGA: Attribute aware Grounded Accuracy

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- AwGA alleviates the representation fragmentation by using only the slots, which are responsible for the output

# Quantitative results

| Metric | mIoU | Acc. | AwGA |
|---|---|---|---|
| DINOSAURv2 | 47.99 | 66.27 | 11.92 |
| FT-DINOSAUR | **59.09** | 61.05 | 13.25 |
| SPOT | 53.76 | 64.08 | 12.81 |
| mFRESA *(ours)* | 56.92 | **67.58** | **13.91** |



AwGA provides a single metric that jointly evaluates how well the object latents localize the object and how useful are these representations.

# Conclusion

- We propose a VLM-based evaluation which is scalable and evaluates diverse OCL properties

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# Conclusion

- We propose a VLM-based evaluation which is scalable and evaluates diverse OCL properties
- Object discovery is poorly correlated with property prediction
- Our proposed AwGA metric evaluates both “*what*” and “*where*” properties of object latents
- Our proposed mFRESA method is a simple multi-feature slot attention method that outperforms previous OCL methods

# Evaluating Object-Centric Models beyond Object Discovery


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 Stefan Roth [1,2]



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Deutsche Forschungsgemeinschaft
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